



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

05-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,B,C,D	A,B,D	B	B,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	31.25	31.25	0.57 to 0.58	387.50	9.20	14.90 to 15.00
SECTION-II (ii)	Q.	7	8	9			
	A.	23	15	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C,D	B,C	A,B,D	B,C,D	A,B,C,D	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.82 to 2.84	5.00	5.00	52.00	4.00	5.00
SECTION-II (ii)	Q.	7	8	9			
	A.	229	18	5			

PART-3 : MATHEMATICS

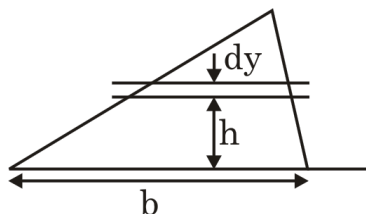
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	A,C,D	B,C	A,B	A,C	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	16.00	512.00	3.00	8.00	12.00	27.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	4	43			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (D)



$$dI = dmy^2 = \frac{m}{\frac{1}{2}bh} \times xdy^2$$

$$\frac{x}{h-y} = \frac{b}{h}$$

$$I = \frac{2mb}{bh^2} \int_0^h (hy^2 - y^3) dy$$

$$= \frac{2m}{h^2} \left[\frac{h^4}{3} - \frac{h^4}{4} \right]$$

$$= \frac{2m}{h^2} \left[\frac{h^4}{12} \right] = \frac{mh^2}{6}$$

2. Ans (B)

$$L_i = L_f$$

$$\frac{2}{5}MR^2\omega_0 = \frac{2}{5}MR^2\omega + m(v + R\omega)R$$

$$\frac{2}{5}MR^2(\omega_0 - \omega) \approx mvR$$

$$\omega_0 - \omega = \frac{5}{2} \frac{mv}{MR}$$

$$\frac{2\pi}{T_0} - \frac{2\pi}{T_0 + \Delta T} = \frac{5}{2} \times \frac{8 \times 10^9 \times 50 \times 1}{6 \times 10^{24} \times 64 \times 10^5}$$

$$\Rightarrow \frac{2\pi}{T_0} \left(\frac{\Delta T}{T_0} \right) = \frac{10^{-17}}{6 \times 64}$$

$$\Delta T = \frac{10^{-17}}{6 \times 6.28 \times 64} \times 8.64 \times 8.64 \times 10^4$$

$$\approx 3 \times 10^{-10} \text{ sec}$$

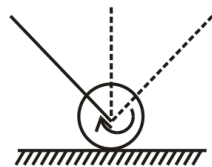
3. Ans (C)

In case 1, energy is lost in jerk when string suddenly becomes taut. But in case 2 there is no sudden jerk.

4. Ans (C)



Spin is a $c\omega$.



Friction $\Rightarrow v_x$ increases

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

$$mg \sin \theta R = 2mR^2\alpha$$

$$mg \sin \theta - f = mR\alpha$$

$$= \frac{mg \sin \theta}{2}$$

$$f = \frac{mg \sin \theta}{1 + \frac{mR^2}{I}} = \frac{mg \sin \theta}{2}$$

$$= \frac{1 \times 10 \times \frac{3}{5}}{2} = 3N$$

$$N = mg \cos \theta = 1 \times 10 \times \frac{4}{5} = 8N$$

6. Ans (A,B,C,D)

$$40 \times 4 \cos 30^\circ = (40 + 20)u_c$$

$$\frac{160}{60} \times \frac{\sqrt{3}}{2} = u_c$$

$$u_c = \frac{4}{\sqrt{3}}$$

$$\frac{1}{2} \times 40 \times \left(\frac{4\sqrt{3}}{2} \right)^2 = \frac{1}{2} \times 60 \times \left(\frac{4}{\sqrt{3}} \right)^2 + \frac{1}{2} \times 6000x^2$$

$$40 \times 4 \times 3 = \frac{60 \times 16}{3} + 6000x^2$$

$$480 \left(1 - \frac{2}{3} \right) = 6000x^2$$

$$x = \sqrt{\frac{160}{6000}} = \sqrt{\frac{2}{75}}$$

After coming back,

$$40 \times \frac{4}{\sqrt{3}} = 40v_p + 20v_c$$

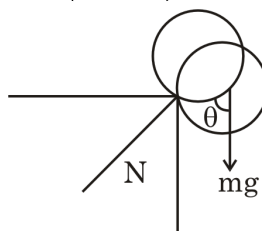
$$e = 1 = \frac{v_c - v_p}{4/\sqrt{3}}$$

$$\frac{8}{\sqrt{3}} = 2v_p + v_c$$

$$-\frac{4}{\sqrt{3}} = -v_c + v_p$$

$$v_p = \frac{4}{3\sqrt{3}}; v_c = \frac{16}{3\sqrt{3}}$$

7. Ans (A,B,D)



$$mg \cos \theta - N = m\omega^2 R$$

$$N = 0$$

$$\omega = \sqrt{\frac{g \cos \theta}{R}} \text{ for both}$$

Case-1 :

$$mg(R - R \cos \theta_1) = \frac{1}{2} \times \frac{3}{2} mR^2 \omega^2$$

Case-2 :

$$mg(R - R \cos \theta_2) = \frac{1}{2} \times mR^2 \omega^2$$

$$\frac{1 - \cos \theta_1}{1 - \cos \theta_2} = 3$$

$$\cos \theta_1 < \cos \theta_2 \Rightarrow Q_1 > Q_2$$

No torque after that $\Rightarrow \omega = \text{const.}$

$$v_x = \omega R \cos \theta = \sqrt{\frac{g \cos^3 \theta}{R}}$$

$$v_{x_1} < v_{x_2}$$

8. Ans (B)

$$200 \times 2 - 200 \times 1 = \frac{1}{2} \times 15 \times 0.1^2 \omega^2$$

$$+ \frac{1}{2} \times 2.0 \times 0.15^2 \omega^2 + \frac{1}{2} \times 20 \times 0.075^2 \omega^2$$

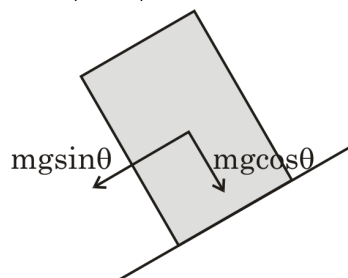
$$\omega^2 = \frac{200 \times 2}{0.15 + 0.45 \times \frac{5}{4}} = \frac{1600 \times 100}{2445}$$

$$\omega = \frac{400}{\sqrt{2445}}$$

$$v_A = \frac{40 \times 0.15}{\sqrt{2445}} = 60$$

$$v_B = \frac{30}{\sqrt{2445}}$$

9. Ans (B,D)



to topple

$$mg \sin \theta \frac{h}{2} > mg \cos \theta \frac{d}{2}$$

$$\tan \theta > \frac{d}{h}$$

10. Ans (A,B,C)

$$(a_G)_t = \alpha r = \alpha(1.5 \text{ m})$$

$$(a_G)_n = \omega^2 r = (5 \text{ rad/s})^2 (1.5 \text{ m}) = 37.5 \text{ m/s}^2$$

$$\Sigma F_t = m(a_G)_t ; 100 \text{ N} = 50 \text{ kg}[\alpha(1.5 \text{ m})]$$

$$\alpha = 1.33 \text{ rad/s}^2$$

$$\Sigma F_n = m(a_G)_n ; T_{AB} + T_{CD} - 50(9.81) \text{ N} = 50 \text{ kg}(37.5 \text{ m/s}^2)$$

$$T_{AB} + T_{CD} = 2365.5$$

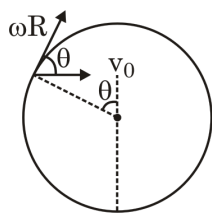
$$\curvearrowright + \Sigma M_G = 0 ; T_{CD}(1 \text{ m}) - T_{AB}(1 \text{ m}) = 0$$

$$T_{AB} = T_{CD} = 1182.75 \text{ N} = 1.18 \text{ kN}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (31.25)



$$v_y = \omega R \sin \theta = v_0 \sin \theta$$

$$h_{\max} = \frac{(v_0 \sin \theta)^2}{2g} + R + R \cos \theta$$

$$\frac{dh}{d\theta} = 0$$

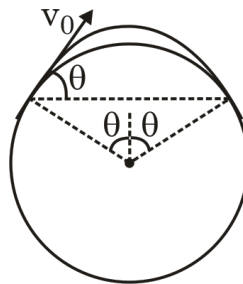
$$= \frac{2v_0^2 \sin \theta \cos \theta}{2g} - R \sin \theta = 0$$

$$\Rightarrow \cos \theta = \frac{gR}{v_0^2} = \frac{1}{4}$$

$$h = \frac{4}{20} \times \frac{15}{16} + 0.1 \times \frac{5}{4} = \frac{15+10}{80} = \frac{25}{80}$$

2. Ans (31.25)

From frame of wheel :-



$$\text{Range} = 2R \sin \theta = \frac{v_0^2 \times 2 \sin \theta \cos \theta}{g}$$

$$\Rightarrow \cos \theta = \frac{Rg}{v_0^2} = \frac{1}{4}$$

$$\Rightarrow \text{same} = 31.25 \text{ cm}$$

3. Ans (0.57 to 0.58)

$$m_1 S_{1x} + m_2 S_{2x} = 0$$

$$20 S_{1x} + 40 S_{2x} = 0$$

$$S_{12} = 2$$

$$S_{12x} = 2 \cos 30^\circ = \sqrt{3}$$

$$S_{1x} - S_{2x} = \sqrt{3}$$

$$\sqrt{3} + 3S_{2x} = 0$$

$$S_{2x} = \frac{-1}{\sqrt{3}} = 0.57 \text{ m}$$

4. Ans (387.50)

$$w_m + w_{gr} - 20 \times 2 \sin 30^\circ \times g = \Delta K$$

$$\Delta K = \frac{1}{2} u v_{\text{rel}}^2 + \frac{1}{2} M v_0^2$$

$$v_{bc} = 5 \text{ m/s}$$

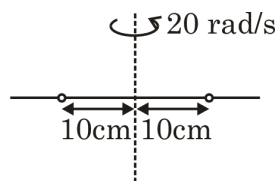
$$v_{by} = v_{by} - v_{cy} = v_{bcy} = 5 \sin 30^\circ = 2.5 \text{ m/s}$$

$$v_{\text{cm}} \text{ in } x \text{ dir.} = 0$$

$$v_{cy} = \frac{20 \times 2.5 + 40 \times 0}{60} = \frac{5}{6}$$

$$w_m = 200 + \frac{1}{2} \times \frac{20 \times 40}{60} \times 5^2 + \frac{1}{2} \times 60 \times \left(\frac{5}{6}\right)^2 = 387.5 \text{ J}$$

5. Ans (9.20)



$$L_i = L_f$$

$$\left(\frac{100 \times 50^2}{12} + 30 \times 10^2 \times 2 \right) \times 20$$

$$= \left(\frac{100 \times 50^2}{12} + 30 \times 25^2 \times 2 \right) \times \omega$$

$$\frac{\frac{250000}{12} + 6000}{\frac{250000}{12} + 37500} \times 20 = \omega$$

$$= \frac{322000}{700000} \times 20$$

$$\omega = 9.2 \text{ rad/s}$$

6. Ans (14.90 to 15.00)

$$\frac{1}{2}mv^2 \times 2 + \frac{1}{2}I\omega^2 = \frac{1}{2}I_0\omega_0^2$$

$$v = \sqrt{\frac{I_0\omega_0^2 - I\omega^2}{2m}}$$

$$= \sqrt{\frac{322000 \times 400 - 700000 \times \left(\frac{46}{5}\right)^2}{2 \times 3}}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (23)

$$0.2N_2 = 1500 a_{\max}$$

$$N_1 + N_2 = 1500 g$$

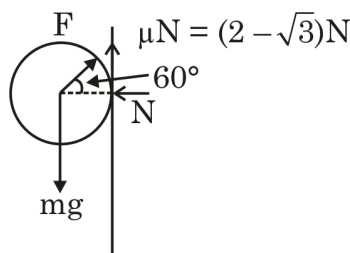
$$N_2 \times 2 - 1500 g \times 0.75 - 1500 a \times 0.35 = 0$$

$$a = \frac{2}{1.544}$$

$$v = at$$

$$t = 23.16 \text{ sec}$$

8. Ans (15)



$$F \cos 60 = N$$

$$F \sin 60 + \mu N = mg$$

$$N = \frac{mg}{2} = 100$$

$$f = (2 - \sqrt{3}) \times 100$$

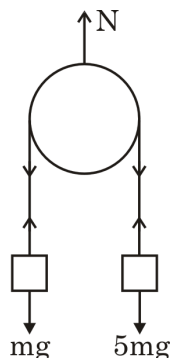
$$\tau = fR = \frac{1}{2} \times 20R^2\alpha$$

$$\alpha = \frac{f}{1.5} = \frac{2}{3} \times (2 - \sqrt{3}) \times 100$$

$$t = \frac{\omega}{\alpha} = \frac{9}{10} (2 + \sqrt{3})$$

$$= \frac{(2 - \sqrt{3}) 1000 \times 3}{2 (2 - \sqrt{3}) \times 100} = 15 \text{ sec}$$

9. Ans (3)



$$5mg - T = 5m \cdot R\alpha$$

$$T + T' + mg = N$$

$$TR - T'R - \mu NR = \frac{1}{2} mR^2\alpha$$

$$T' - 2mg = 2mR\alpha$$

$$T - T' - 0.1(T + T' + mg) = \frac{1}{2} ma$$

$$0.9T - 1.1T' - 0.1mg = \frac{1}{2} ma$$

$$T' = 2mg + 2ma$$

$$T = 5mg - 5ma$$

$$4.5 mg - 4.5 ma - 2.5 mg - 2.2 ma - 0.1 mg = 0.5 ma$$

$$2.2 mg = 7.2 ma$$

$$a = \frac{220}{72} \approx 3 \text{ m/s}^2$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

$$D_1 = |10|, D_2 = \begin{vmatrix} 10 & 3 \\ 3 & 10 \end{vmatrix} = 10 \times 10 - 3 \cdot 3 = 91$$

$$D_3 = \begin{vmatrix} 10 & 3 & 0 \\ 3 & 10 & 3 \\ 0 & 3 & 10 \end{vmatrix} = 10 \begin{vmatrix} 10 & 3 \\ 3 & 10 \end{vmatrix} - 3 \begin{vmatrix} 3 & 3 \\ 0 & 10 \end{vmatrix} + 0 \begin{vmatrix} 3 & 10 \\ 0 & 3 \end{vmatrix}$$

$$D_n = 10D_{n-1} - 9D_{n-2} = 10(D_{n-1} - D_{n-2})$$

$$+ D_{n-2} \Rightarrow D_n - D_{n-1} = 9(D_{n-1} - D_{n-2})$$

$$D_2 = 10D_1 - 9D_0 \Rightarrow D_0 = 1 \text{ by recursion}$$

$$\text{formula } D_4 = 7381$$

$$D_1 = D_0 = 10 - 1 = 9$$

$$D_2 - D_1 = 81 = 9^2$$

$$D_3 - D_2 = 820 - 91 = 729 = 9^3$$

$$D_4 - D_3 = 7381 - 820 = 6561 = 9^4$$

$$D_n = D_0 + 9^1 + 9^2 + \dots + 9^n = \sum_{i=0}^n 9^i$$

Desired sum

$$\sum_{n=1}^{\infty} \frac{1}{\frac{8 \cdot 9^{n+1} - 1}{8} + 1} = \sum_{n=1}^{\infty} \frac{1}{9^{n+1} - 1 + 1} = \sum_{n=1}^{\infty} \frac{1}{9^{n+1}} = \frac{1}{72}$$

2. Ans (A)

$$P = \begin{bmatrix} \omega^2 & \omega^3 & \omega^4 & \dots & \omega^{n+2} \\ \omega^3 & \omega^4 & \omega^5 & \dots & \omega^{n+3} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \omega^{n+2} & \omega^{n+3} & \dots & \dots & \omega^{2n+4} \end{bmatrix}$$

$$P^2 = \begin{bmatrix} \omega^4 + \omega^6 & \dots & \omega^5 + \omega^7 & \dots & \dots \\ \omega^5 + \omega^7 & \dots & \dots & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \omega^{n+4} + \omega^{n+6} & \dots & \dots & \dots & \omega^{2n+4} + \omega^{2n+6} \end{bmatrix}$$

= Null matrix if n is a multiple of 3.

3. Ans (A)

$$\sum_{k=0}^{n+1} \sum_{j=1}^{k+1} \left(\sum_{i=1}^k {}^{i+1}C_j - {}^iC_j \right) = \sum_{k=0}^{n+1} \sum_{j=1}^{k+1} ({}^{k+1}C_j - {}^1C_j)$$

$$= \sum_{k=0}^{n+1} (2^{k+1} - 2)$$

$$= \sum_{k=0}^{n+1} 2^{k+1} - 2 \sum_{k=0}^{n+1} 1 = 2 \cdot (2^{n+2} - 1) - 2(n+2) = 2^{n+3} - 2n - 6$$

4. Ans (A)

$$D_1 = 2\cos x$$

$$D_2 = \begin{vmatrix} 2\cos x & 1 \\ 1 & 2\cos x \end{vmatrix} = 4\cos^2 x - 1$$

$$= 2(\cos 2x + 1) - 1 = 2\cos 2x + 1$$

$$D_3 = \begin{vmatrix} 2\cos x & 1 & 0 \\ 1 & 2\cos x & 1 \\ 0 & 1 & 2\cos x \end{vmatrix}$$

$$= 2\cos x(4\cos^2 x - 1) - 1(2\cos x)$$

$$= 2\cos x(4\cos^2 x - 2) = 4\cos x \cos 2x$$

$$D_3 - 2D_2\cos x + D_1 = 0$$

$$4\cos x \cos 2x - 2(2\cos 2x + 1)\cos x + 2\cos x = 0$$

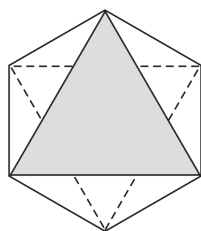
$$0 = 0$$

PART-3 : MATHEMATICS

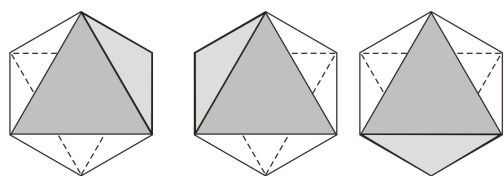
SECTION-I (ii)

5. Ans (A,B,D)

Since the octahedron is indistinguishable by rotations, without loss of generality fix a face to be red.



There are $7!$ ways to arrange the remaining seven colors, but there still are three possible rotations about the fixed face, so the answer is $7!/3 = 1680$



6. Ans (A,C,D)

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad a \neq b \neq c \neq d > 0$$

$$A + A^T = \begin{pmatrix} 2a & b+c \\ b+c & 2d \end{pmatrix}; |A + A^T|$$

$$= 4ad - (b+c)^2 = 0 \Rightarrow b+c = 2\sqrt{ad}$$

$$\Rightarrow b+c = 2\sqrt{ad} > 2\sqrt{bc} \quad (\text{A.M} > \text{G.M})$$

$$\Rightarrow ad > bc \Rightarrow ad - bc > 0 \quad (\text{as } a \neq b \neq c \neq d > 0)$$

$$\Rightarrow \det A > 0$$

$$|A - A^T| = \left| \begin{pmatrix} 0 & b-c \\ c-b & 0 \end{pmatrix} \right| = 0 + (b-c)^2 > 0;$$

$$|AA^T| = |A||A^T| = |A|^2 = (\det A)^2 > 0$$

7. Ans (B,C)

$$3ABA^{-1} + A = 2A^{-1}BA$$

$$\Rightarrow 3ABA^{-1} + A + 2A = 2A^{-1}BA + 2A$$

$$\Rightarrow 3A(BA^{-1} + I) = 2(A^{-1}B + I)A$$

$$\Rightarrow 3A(B + IA)A^{-1} = 2A^{-1}(B + AI)A$$

$$\Rightarrow 3A(B + A)A^{-1} = 2A^{-1}(B + A)A$$

$$\text{Let } B + A = X$$

$$\Rightarrow 3AXA^{-1} = 2A^{-1}XA \Rightarrow 3^n |A||X||A^{-1}|$$

$$= 2^n |A^{-1}||X||A| \Rightarrow 3^n |X|$$

$$= 2^n |X| \quad (\text{as } |A| \neq 0)$$

$$\Rightarrow |X| = 0 \text{ or } |A+B| = 0$$

$$\text{Let } M = ABA^{-1} - A^{-1}BA$$

$$\therefore AM = A^2BA^{-1} - BA \Rightarrow BA = A^2BA^{-1} - AM$$

$$\text{Now } 3ABA^{-1} + A = 2A^{-1}BA$$

$$= 2A^{-1}(A^2BA^{-1} - AM) = 2ABA^{-1} - 2M$$

$$\Rightarrow ABA^{-1} + A = -2M$$

$$\Rightarrow A(BA^{-1} + I) = -2M$$

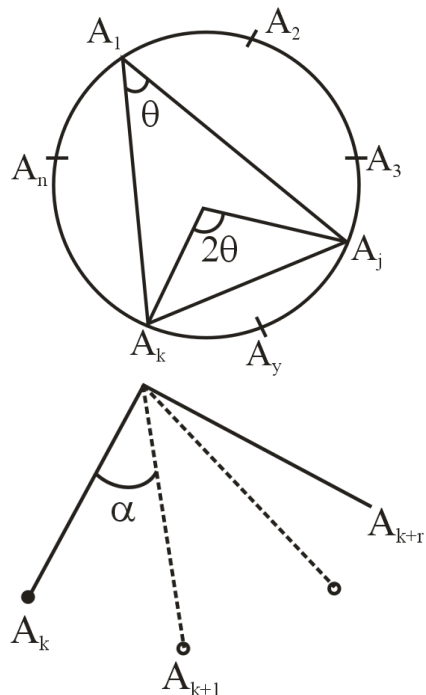
$$A(A+B)A^{-1} = -2M$$

Taking determinants both sides we get

$$|-2M| = |A||A+B||A^{-1}| = 0$$

$$\Rightarrow |ABA^{-1} - AB^{-1}A| = 0$$

8. Ans (A,B)



$$\alpha = 2\pi/n$$

Suppose $\Delta A_i A_j A_k$ is obtuse at A_1 .

Number of vertices between A_i & A_j , A_j & A_k and

A_k and A_1 respectively are x, y & z

$$x + y + z = n - 3 \quad \dots(1)$$

$$\text{since } 90^\circ < \theta \Rightarrow 180^\circ \leq 2\theta$$

2θ = Angle subtended by 'y + z'

vertices at centre

$$2\theta = (y + 1) \cdot \frac{2\pi}{n} > \pi$$

$$y + 1 > \frac{n}{2}$$

$$y > \frac{n}{2} - 1 \Rightarrow y \geq \frac{n}{2}$$

Therefore we need to find the solution of

$x + y + z = n - 3$ under the condition $x, z \geq 0$

and $y \geq \frac{n}{2}$

coeff. x^{n-3} in

$$(x^0 + x^1 + x^2 + \dots)^2 \left(x^{\frac{n}{2}} + x^{\frac{n}{2}+1} + x^{\frac{n}{2}+2} + \dots \right)$$

$$= x^{\frac{n}{2}} (1 - x)^{-3}$$

$$n = 2k$$

coeff. of x^{2k-3} in $x^k (1 - x)^{-3}$

$$\Rightarrow {}^{3+k-3-1}C_{k-3} = {}^{k-1}C_{k-3} = {}^{k-1}C_2 = \frac{(k-1)(k-2)}{2}$$

this in number of obtuse triangle when A_1 is obtuse similarly other vertices can be obtuse

$$= \frac{2k(k-1)(k-2)}{2}$$

$$= k(k-1)(k-2) = \frac{n(n-2)(n-4)}{8}$$

$$n = 2k + 1, x, z \geq 0 \text{ and}$$

$$y + 1 > \frac{2k+1}{2} \Rightarrow y > k - \frac{1}{2} \Rightarrow y \geq k$$

coeff. of x^{n-3} in $(x^0 + x^1 + x^2 + \dots)$

$$w = y - k \Rightarrow y = W + k, W \geq 0$$

$$x + y + z = n - 3 \Rightarrow x + y + w + k = 2k - 2$$

$$x + w + z = k - 2 \Rightarrow {}^{k-2+3-1}C_{3-1} = {}^kC_2$$

$$\frac{k(k-1)}{2}$$

Total triangles

$$= \frac{n}{2} \cdot \frac{(n-1)}{2} \cdot \frac{(n-3)}{2} = \frac{n(n-1)(n-3)}{8}$$

option A and B are correct

9. Ans (A,C)

$$1, 4, 7, 10 \rightarrow 1, 1, 1, 1$$

$$2, 5, 8, 11 \rightarrow 2, 2, 2, 2$$

$$3, 6, 9, 12 \rightarrow 0, 0, 0, 0$$

	C_1	C_2	C_3
Possible columns	0	0	0
	1	2	2

There will $2C_1, 2C_2, 2C_3$

C_1	C_1	C_2	C_2	C_3	C_3
0	1	0	2	1	2
1	0	2	0	2	1

or

1	0	2	0	2	1
0	1	0	2	1	2

If we fix one column other 5 columns will have only one arrangement :

One column can be fixed in 2 ways.

$$\text{Hence total arrangement} = \frac{6!}{2! 2! 2!} \times 2$$

Now 0's are actual 3, 6, 9, 12 which can be arranged in $4!$ ways. similarly 1, 1, 1, 1 and 2, 2, 2, 2 can also be arranged in $4!$ ways.

$$\text{Therefore, } N = \frac{6!}{2! 2! 2!} \times 2! \times 4! \times 4! \times 4!$$

$$N = 2^{11} \cdot 3^5 \cdot 5$$

$$\text{Total divisors} = 12 \times 6 \times 2 = 144$$

10. **Ans (A, B, C)**

$$\Rightarrow ({}^{29}C_4 + {}^{29}C_5) - ({}^{29}C_5 + {}^{29}C_6) + \dots - ({}^{29}C_{27} + {}^{29}C_{28})$$

$$= {}^{29}C_4 - 29$$

$$\Rightarrow 3P = {}^{29}C_4 \Rightarrow P = 3 \times 7 \times 13 \times 29$$

$${}^QC_R = {}^{104}C_{94} \Rightarrow Q = 104, R = 94 \text{ or } 10$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (16.00)**

$$A + B^T = \text{adj}B \Rightarrow A^T + B = (\text{adj}B)^T \text{ and}$$

$$B + A^T = \text{adj}A$$

$$\Rightarrow (\text{adj}B)^T = \text{adj}A \Rightarrow |B|^2 = |A|^2$$

$$\Rightarrow |B| = \pm |A|$$

$$(|B|B^{-1})^T = |A|A^{-1}$$

$$\text{If } |B| = |A|$$

$$\Rightarrow (B^{-1})^T = A^{-1} = (B^T)^{-1} \Rightarrow A = B^T$$

$$\therefore \text{adj}B = 2B^T \Rightarrow |B|^2 = 2^3 |B|$$

$$\Rightarrow |B| = 8, |A| = 8$$

$$\text{If } |B| = -|A| \Rightarrow (-B^{-1})^T = A^{-1} = (-B^T)^{-1}$$

$$\Rightarrow A = -B^T$$

$$\therefore \text{adj}B = 0 \text{ which is impossible}$$

$$\text{Now, } A = B^T$$

$$\text{adj}B = 2A$$

$$(\text{adj}B)B = 2AB = |B|^T \Rightarrow AB = 4I$$

$$\text{Also } B(\text{adj}B) = 2BA = |B|I \Rightarrow BA = 4I$$

2. **Ans (512.00)**

$$A + B^T = \text{adj}B \Rightarrow A^T + B = (\text{adj}B)^T \text{ and}$$

$$B + A^T = \text{adj}A$$

$$\Rightarrow (\text{adj}B)^T = \text{adj}A \Rightarrow |B|^2 = |A|^2$$

$$\Rightarrow |B| = \pm |A|$$

$$(|B|B^{-1})^T = |A|A^{-1}$$

$$\text{If } |B| = |A|$$

$$\Rightarrow (B^{-1})^T = A^{-1} = (B^T)^{-1} \Rightarrow A = B^T$$

$$\therefore \text{adj}B = 2B^T \Rightarrow |B|^2 = 2^3 |B|$$

$$\Rightarrow |B| = 8, |A| = 8$$

$$\text{If } |B| = -|A| \Rightarrow (-B^{-1})^T = A^{-1} = (-B^T)^{-1}$$

$$\Rightarrow A = -B^T$$

$$\therefore \text{adj}B = 0 \text{ which is impossible}$$

$$\text{Now, } A = B^T$$

$$\text{adj}B = 2A$$

$$(\text{adj}B)B = 2AB = |B|^T \Rightarrow AB = 4I$$

$$\text{Also } B(\text{adj}B) = 2BA = |B|I \Rightarrow BA = 4I$$

$$|AB + BA| = |8I| = 8^3 = 2^9 = 512$$

3. **Ans (3.00)**

$$\text{We have, } (1 + px + x^2)^n = 1 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{2n}x^{2n}$$

$$\text{On replacing } x \text{ by } x^4, \text{ we get}$$

$$(1 + px^4 + x^8)^n = 1 + a_1x^4 + a_2x^8 + a_3x^{12} + \dots + a_{2n}x^{8n}$$

$$\text{On dividing by } x^3 \text{ both side, we get}$$

$$\frac{(1 + px^4 + x^8)^n}{a_{2n}x^{8n-3}x^3} = \frac{1}{x^3} + a_1x + a_2x^5 + a_3x^9 + \dots +$$

$$[\text{on differentiate w.r.t } x, \text{ we get}]$$

$$\frac{nx^3(4px^3 + 8x^7)(1 + px^4 + x^8)^{n-1} - 3x^2(1 + px^4 + x^8)^n}{x^6}$$

$$= \frac{-3}{x^4} + a_1 + 5a_2x^4 + 9a_3x^8 + \dots + (8n-3)a_{2n}x^{8n-4}$$

$$\text{Put } x = 1 \text{ both side, we get}$$

$$n(4p+8)(2+p)^{n-1} - 3(2+p)^n$$

$$= -3 + a_1 + 5a_2 + 9a_3 + \dots + (8n-3)a_{2n}$$

$$(p+2)^n \cdot (4n-3) + 3 = a_1 + 5a_2 + 9a_3 + \dots +$$

$$(8n-3)a_{2n}$$

$$\text{When } a_1 + 5a_2 + 9a_3 + \dots + (8n-3)a_{2n} \text{ is divide by}$$

$$(p+2) \text{ the remainder is } 3.$$

4. Ans (8.00)

We have, $(1 + px + x^2)^n = 1 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{2n}x^{2n}$

On replacing x by x^2 , we get

$$(1 + px^2 + x^4)^n = 1 + a_1x^2 + a_2x^4 + a_3x^6 + \dots + a_{2n}x^{4n}$$

On dividing by x , we get

$$\frac{(1 + px^2 + x^4)^n}{x} = \frac{1}{x} + a_1x + a_2x^3 + a_3x^5 + \dots + a_{2n}x^{4n-1}$$

On differentiating w.r.t. to x , we get

$$\frac{n(2px + 4x^3)(1 + px^2 + x^4)^{n-1} - (1 + px^2 + x^4)^n}{x^2} = \frac{-1}{x^2} + a_1 + 3a_2x^2 + 5a_3x^4 + \dots + (4n-1)x^{4n-2} \cdot a_{2n}$$

Put $p = -3$ and $x = 1$

$$\begin{aligned} &\Rightarrow n(-6+4)(1-3+1)^{n-1} - (1-3+1)^n \\ &= -1 + a_1 + 3a_2 + 5a_3 + \dots + (4n-1)a_{2n} \\ &\Rightarrow -2n(-1)^{n-1} - (-1)^n + 1 = a_1 + 3a_2 + 5a_3 + \dots + (4n-1)a_{2n} \\ &\Rightarrow 2n(-1)^n - (-1)^n + 1 = a_1 + 3a_2 + 5a_3 + \dots + (4n-1)a_{2n} \\ &\Rightarrow 2n - 1 + 1 = a_1 + 3a_2 + 5a_3 + \dots + (4n-1)a_{2n} \\ &\Rightarrow a_1 + 3a_2 + 5a_3 + \dots + (4n-1)a_{2n} \\ &= 2n [\because (-1)^n = 1, n \text{ is even}] \end{aligned}$$

5. Ans (12.00)

Let's count the Δ 's right way

FS	SS	TS
1	1	1
2	2	1,2,3
3	3	1,2,3,4
4	4	1,2,3,4

FS $\rightarrow$ First side, SS $\rightarrow$ Second side

TS $\rightarrow$ Third side

$$\therefore \text{Total number of ways} = 1 + 3 + 4 + 4 = 12$$

6. Ans (27.00)

1	1	1
2	2	1,2,3
3	3	1,2,3,4,5
4	4	1,2,3,4,5,6
5	5	1,2,3,4,5,6
6	6	1,2,3,4,5,6

$$= 1 + 3 + 5 + 6 + 6 + 6 = 27$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (7)

Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Note that $S_i \cap S_j = S_i$ is equivalent to $S_i \subseteq S_j$. We use constructive counting and select S_5 first, and then move down to S_1 . First, note that there are $\binom{7}{5}$ ways to choose S_5 . Now, S_4 must be a subset of S_5 (as if $S_4 \cap S_5 = \phi$ then $|S| \geq 4 + 5 = 9$), so there are $\binom{5}{4}$ ways to choose S_4 . Moreover, if S_3 is not a subset of S_4 , then $S_3 = S - S_4$. But then $S_3 \cap S_5 \neq S_3$ and $S_3 \cap S_5 \neq \phi$, a contradiction. Thus $S_3 \subseteq S_4 \subseteq S_5$ and it follows that there are $\binom{4}{3}$ ways to choose S_3 . Now, either S_2 is a subset of S_3 , or $S_2 \cap S_3 = \phi$. In the first case, there are $\binom{3}{2}$ ways to choose S_2 . In the latter case, since S_2 is not a subset of S_3 , it cannot be a subset of S_4 and S_5 . It follows that $S_2 \cap S_4, S_2 \cap S_5 = \phi$ and $S_2 = S - S_5$. Thus, there are $\binom{3}{2} + 1$ total ways to select S_2 . Finally, note that S_1 can be any of $\{1, 2, 3, 4, 5, 6, 7\}$. The answer is

$$\binom{7}{5} \binom{5}{4} \binom{4}{3} \left(\binom{3}{2} + 1 \right) 7 = 21 \cdot 5 \cdot 4 \cdot 4 \cdot 7 = 11760$$

8. Ans (4)

$$\begin{aligned}
 p &= \sum_{r=1}^{50} (-1)^{r-1} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r} \right) \cdot {}^{50}C_r \\
 &= \int_0^1 \sum_{r=1}^{50} (-1)^{r-1} (1 + x + x^2 + \dots + x^{r-1}) \cdot {}^{50}C_r \\
 &= \int_0^1 \sum_{r=1}^{50} (-1)^{r-1} \cdot {}^{50}C_r \cdot \frac{(1-x^r)}{1-x} dx \\
 &= \int_0^1 \frac{1}{1-x} \sum_{r=1}^{50} \left\{ (-1)^{r-1} \cdot {}^{50}C_r - (-1)^{r-1} \cdot {}^{50}C_r \right\} dx \\
 &= \int_0^1 \frac{1}{1-x} \left\{ 1 + ((1-x)^{50} - 1) \right\} dx \\
 &= \int_0^1 (1-x)^{49} dx = \frac{1}{50} = p \Rightarrow GE = 4
 \end{aligned}$$

9. Ans (43)

$$\text{Tr}(A) = 16, \text{Det}(A) = -17$$

$$a + d = 16 \text{ and } ad - bc = -17$$

 Given $a < b < c < d$

$$(a, d) = (1, 15)(2, 14)(3, 13)(4, 12)(5, 11)(6, 10)(7, 9)$$

$$bc = ad + 17 = 32, 45, 56, 65, 72, 77, 80$$

$$(b, c) = (4, 8)(5, 9)(7, 8)(5, 13)(8, 9)(7, 11)(8, 10)$$

$$(a, b, c, d) = (1, 4, 8, 15)(2, 5, 9, 14)(3, 7, 8, 13)$$

$$(4, 5, 13, 12)(5, 8, 9, 11)(6, 7, 11, 10)(7, 8, 10, 9)$$

Possible values are (2, 5, 9, 14) (3, 7, 8, 13) (5, 8, 9, 11)

Exactly two of a, b, c, d are prime and pair wise also coprime

Case-1 : (2, 5, 9, 14) 2 primes but pair wise not coprime

 Case-2 : (3, 7, 8, 13) 3 primes and pair wise coprime
 (given exactly two are primes)

Case-2 : (5, 8, 9, 11) 2 primes and pair wise coprime

Only possible answer is (5, 8, 9, 11)

$$|B| = bd - ac = 88 - 45 = 43$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

05-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	B,C,D	A,B,C,D	A,C,D	B,C,D	B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	80.00	71.00	0.75	43.75	60.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	450	84	260			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B,D	A,B	A,B,C	A,B,C,D	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.73	1400.00	4.00	150.00	6.00	9.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	6	17			

PART-3 : MATHEMATICS

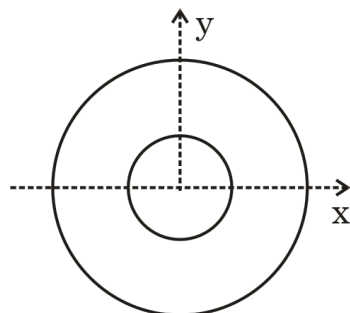
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B,C	A,B,D	A,B	B,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	D	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.50	1.20	56.00	5.00	0.50	24.00
SECTION-II (ii)	Q.	7	8	9			
	A.	9	1	0			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)



$$2I_Z = \frac{1}{2} \times 2M (R_1^2 + R_2^2)$$

$$I_Z = \frac{1}{2} M (R_1^2 + R_2^2)$$

$$= I_x + I_y \text{ or } I'_x + I'_y$$

$$I'_x = I'_y \Rightarrow I_x = I_y = I'_x = I'_y$$

2. Ans (B,C,D)

S_y same $v_{yB} = -ve$

$$4 \times 600 = 2.5 \times v_B \frac{\sqrt{3}}{2}$$

$$-1.5 \frac{v_A}{\sqrt{2}}$$

$$1.5 \times \frac{v_A}{\sqrt{2}} = \frac{2.5v_B}{2}$$

$$v_A = \frac{5v_B}{3\sqrt{2}}$$

$$2400 = \frac{5\sqrt{3}}{4} v_B - \frac{3}{2\sqrt{2}} \times \frac{5v_B}{3\sqrt{2}}$$

$$= \frac{5}{4} (\sqrt{3} - 1) v_B$$

$$v_B = 960 (\sqrt{3} + 1) \text{ m/s}$$

$$v_A = \frac{5}{3\sqrt{2}} \times 960 (\sqrt{3} + 1) = 800 (\sqrt{6} + \sqrt{2})$$

3. Ans (A,B,C,D)

$mg = 1\text{N}$ so till 0.8 sec, it speeds up.

$$J + J_{gr} = \Delta p$$

$$\frac{1}{2} \times 0.2 \times 13 + \frac{1}{2} \times 0.1 \times 18 + 0.5 \times 5 - 1 \times 0.8 = 0.1 v$$

$$v = 39 \text{ m/s}$$

$$t = \frac{39}{10} = 3.9 \text{ sec}$$

4. Ans (A,C,D)

Case-I

$$mg - F = mR\alpha$$

$$FR = \frac{1}{2} MR^2 \alpha$$

$$mg = \left(m + \frac{1}{2} M \right) R\alpha$$

$$R\alpha = \frac{mg}{m + \frac{1}{2} M}$$

$$F = \frac{1}{2} \frac{Mmg}{m + \frac{1}{2} M}$$

$$\Rightarrow \frac{Mmg}{m + \frac{M}{2}} = \frac{2Mg}{3}$$

$$3m = 2m + M$$

$$\Rightarrow m = M$$

$$\alpha = \alpha' \Rightarrow \omega \text{ same}$$

Case-2

$$MgR = \frac{3}{2} MR^2 \alpha'$$

$$R\alpha' = \frac{2g}{3}$$

$$Mg - F = MR\alpha'$$

$$F = \frac{Mg}{3}$$

5. Ans (B,C,D)

$$N = 0$$

$$mg \cos \theta = \frac{mv^2}{R}$$

$$mg(R - R \cos \theta) = \frac{1}{2}mv^2 + \frac{1}{2} \frac{Iv^2}{r^2}$$

$$v^2 = \frac{2gR(1 - \cos \theta)}{1 + \frac{k^2}{r^2}} = gR \cos \theta$$

$$2 \sec \theta = 3 + \frac{k^2}{r^2}$$

$$k_{\text{sphere}} < k_{\text{ring}}$$

$$\sec \theta_S < \sec \theta_R$$

$$\Rightarrow \theta_S < \theta_R$$

$$\Rightarrow v_S > v_R$$

6. Ans (B,C)

$$v_x = 10 \cos 53 = 6 \text{ m/s}$$

$$t_1 = \frac{3}{4x} = 0.5 \text{ sec}$$

$$T = \frac{24y}{9} = 1.6 \text{ sec.}$$

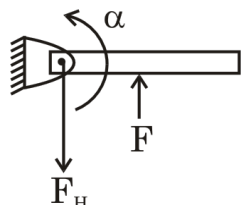
$$t_2 = 1.1 \text{ sec}$$

$$v_x = \frac{3}{1.1} \text{ m/s} \quad e = \frac{v_x}{u_x}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

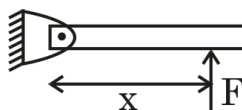


$$\frac{Fw}{3} = \frac{mw^2}{3} \alpha$$

$$F - F_H = ma_c = m \frac{w\alpha}{2} = \frac{F}{2}$$

$$F_H = \frac{F}{2}$$

8. Ans (C)



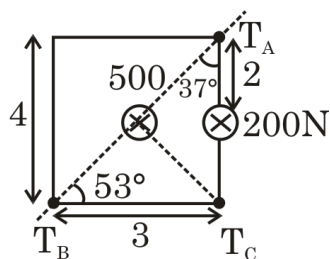
$$F_x = \frac{mw^2}{3} \alpha$$

$$F = ma_c = \frac{mw}{2} \alpha$$

$$m \frac{w}{2} \alpha x = \frac{mw^2}{3} \alpha$$

$$x = \frac{2w}{3}$$

9. Ans (A)

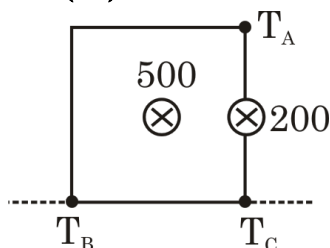


τ about diagonal

$$T_C \times 3 \sin 53 = 200 \times 2 \sin 37^\circ$$

$$T_C = \frac{200 \times 2 \times \frac{3}{5}}{3 \times \frac{4}{5}} = 100 \text{ N}$$

10. Ans (C)



$$T_A + T_B + T_C = 700 \text{ N}$$

$$\Rightarrow T_A + T_B = 600 \text{ N}$$

τ about lower edge

$$(200 + 500) \times 2 = T_A \times 4$$

$$T_A = 350$$

$$T_B = 250$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (71.00)

$$\begin{array}{c} \text{1kg} \quad \text{4m} \quad \text{2m/s} \\ \leftarrow \text{1m/s} \quad \text{4kg} \rightarrow \\ -\frac{dU}{ds} = -5s \Rightarrow U = \frac{5s^2}{2} \end{array}$$

$$s_{\max} \Rightarrow v_{\text{rel}} = 0$$

$$\frac{1}{2} \mu u_{\text{rel}}^2 + u_i = \frac{1}{2} \mu v_{\text{rel}}^2 + u_f$$

$$\frac{1}{2} \times \frac{1 \times 4}{1+4} \times S^2 + \frac{5}{2} \times 4^2 = 0 + \frac{5}{2} \times S^2$$

$$S^2 = 20 \Rightarrow S = 2\sqrt{5} = 4.46 \text{ to } 4.47$$

$$1 \times 0 + 4 \times v = 1 \times -1 + 4 \times 4 \Rightarrow v = \frac{15}{4} \text{ m/s}$$

$$\begin{aligned} \frac{1}{2} \times 1 \times 0^2 + \frac{1}{2} \times 4 \times \frac{15^2}{4} + \frac{5}{2} \times S^2 \\ = \frac{1}{2} \times 1 \times 1^2 + \frac{1}{2} \times 4 \times 4^2 + \frac{5}{2} \times 4^2 \end{aligned}$$

4. Ans (43.75)

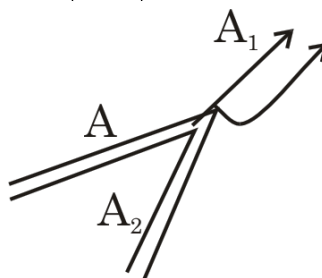
$$e = \sqrt{\frac{h_1}{h_2}} = \frac{3}{4} = 0.75$$

$$\% = \frac{\frac{1}{2} \mu u_{\text{rel}}^2 - \frac{1}{2} \times \mu v_{\text{rel}}^2}{\frac{1}{2} \times m \times u^2 \times 2} =$$

$$\frac{\frac{1}{2} \times \frac{m}{2} (2u)^2 \left(1 - \frac{9}{16}\right)}{\mu u^2}$$

$$= \frac{7}{16} \times 100 = 43.75$$

6. Ans (3.00)



$$Av = A_1v + A_2v$$

$$\frac{dp_1}{dt} = \frac{dm}{dt} v = \rho A_1 v^2$$

$$\frac{dp_2}{dt} = \rho A_2 v^2$$

$$\rho A_1 v^2 - \rho A_2 v^2 = \rho A v^2 \cos \theta$$

$$\Rightarrow A_1 - A_2 = A \cos \theta$$

$$A_1 + A_2 = A$$

$$A_1 = \frac{A}{2} (1 + \cos \theta)$$

$$A_2 = \frac{A}{2} (1 - \cos \theta)$$

$$\frac{Q_1}{Q_2} = \frac{A_1}{A_2} = 3$$

$$F = \rho A v^2 \sin \theta$$

$$= 10^3 \times \sqrt{3} \times 10^{-4} \times 400 \times \frac{\sqrt{3}}{2}$$

$$= 60 \text{ N}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (450)

$$\begin{aligned} x_C &= \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\lambda_0 \int_0^L x dx + \int_0^L \frac{x^3 dx}{L^2}}{\lambda_0 \int_0^L dx + \int_0^L \frac{x^2 dx}{L^2}} \\ &= \frac{\frac{L^2}{2} + \frac{L^2}{4}}{L + \frac{L}{3}} = \frac{\frac{3L^2}{4}}{\frac{4L}{3}} = \frac{9}{16} \times 8 \end{aligned}$$

$$= 4.5 \text{ m}$$

8. Ans (84)

$$L_i = L_f$$

$$\frac{mr^2}{2} \omega_0 = \left(\frac{mr^2}{2} + m\ell^2 \right) \omega$$

$$mg \times 2\ell = 0 - \frac{1}{2} \times \left(\frac{mr^2}{2} + m\ell^2 \right) \omega^2$$

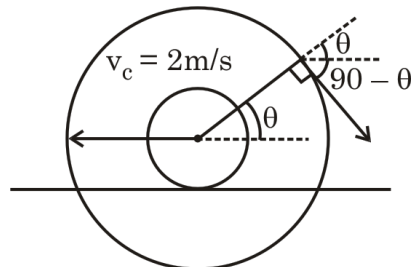
$$\omega = \frac{\omega_0}{1 + \frac{2\ell^2}{r^2}} = \sqrt{\frac{g\ell/r^2}{1 + \frac{2\ell^2}{r^2}}}$$

$$\omega_0 = \sqrt{\frac{8g\ell}{r^2} \left(1 + \frac{2\ell^2}{r^2} \right)}$$

$$= \sqrt{\frac{8 \times 9.8 \times 0.4}{0.2 \times 0.2} (1 + 2 \times 4)}$$

$$= 7 \times 3 \times 4 = 84 \text{ rad/s}$$

9. Ans (260)



$$5 \times 0.52 = 2.6 \text{ m/s} = v_{AC}$$

$$v_{Ay} = 2.6 \cos \theta = 2.6 \times \frac{12}{13} = 2.4$$

$$v_{Ax} = 2.6 \sin \theta - 2$$

$$= 2.6 \times \frac{5}{13} - 2 = -1$$

$$v_A = \sqrt{v_{Ax}^2 + v_{Ay}^2}$$

$$= \sqrt{6.76} = 2.6 \text{ m/s}$$

$$= 260 \text{ cm/s}$$

PART-2 : CHEMISTRY

SECTION-II (i)

2. Ans (1400.00)

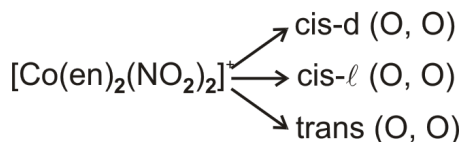
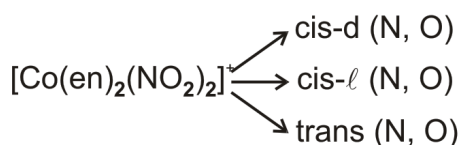
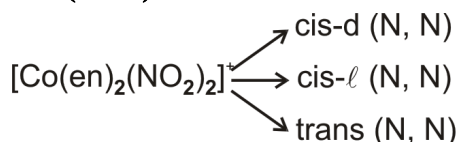
$$\text{CFSE} = -460 - (-436) = -24 \text{ k cal mol}^{-1}$$

$$= -24 \times 350 \text{ cm}^{-1}$$

$$\text{Cr}^{2+} = 3d^4 \quad t_{2g}^{1,1,1} \quad e_g^{1,0}$$

$$-0.6\Delta_0 = -24 \times 350 \Rightarrow \Delta_0 = 14000 \text{ cm}^{-1}$$

6. Ans (9.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C,D)

Here $A = \text{diag}(x + xy + xyz, y + yz + xyz, z + zx + xyz + xyz)$

$$A^{-1} = \text{diag}$$

$$\left(\frac{1}{x + xy + xyz}, \frac{1}{y + yz + xyz}, \frac{1}{z + zx + xyz} \right)$$

and $xyz = 1$

$$\begin{aligned} \text{Tr} A^{-1} &= \frac{1}{x + xy + xyz} + \frac{1}{y + yz + xyz} + \frac{1}{z + zx + xyz} \\ &= \frac{1}{1 + x + xy} + \frac{1}{y + yz + 1} + \frac{1}{1 + zx + z} \\ &= \frac{1}{1 + x + xy} + \frac{1}{x} + \frac{1}{1 + xz + z} \\ &= \frac{1}{1 + x} + \frac{y}{y + 1 + zy} \\ &= \frac{1 + x + xy}{1 + x + xy} + \frac{xy}{xy + x + xyz} = \frac{1 + x + xy}{1 + x + xy} = 1 \end{aligned}$$

Sum of maximum and minimum value of $\text{Tr}(A^{-1})$

$$= 1 + 1 = \frac{2}{1}$$

$$p = 2, q = 1$$

2. **Ans (A,B,C)**

D_n = de arrangement of n persons

$$K = {}^6P_5 (D_6 + D_5) = 6!(265 + 44) = 309 \times 6!$$

3. **Ans (A,B,D)**

Coefficient of

$$(1) x^{17} \rightarrow x^2 \cdot x^5 \cdot x^{10} \rightarrow 1$$

$$(2) x^{16} \rightarrow \left. \begin{array}{l} x^2 x^4 x^{10} \rightarrow 5 \\ x x^5 x^{10} \rightarrow 1 \end{array} \right\} 6$$

$$(3) x^{15} \rightarrow \left. \begin{array}{l} 1 x^5 x^{10} \rightarrow 1 \\ x x^4 x^{10} \rightarrow 5 \\ x^2 x^3 x^{10} \rightarrow 10 \\ x^2 x^5 x^8 \rightarrow 5 \end{array} \right\} 21$$

$$(4) x \rightarrow \left. \begin{array}{l} 1.x.1 \rightarrow 5 \\ x.1.1 \rightarrow 1 \end{array} \right\} 6$$

4. **Ans (A,B)**

$$\text{trace}(B) = \sum_{i=1}^n (2+i) = 2n + \frac{n(n+1)}{2}$$

$$\sum_{n=1}^m \text{trace}(B) = \frac{2m(m+1)}{2} + \frac{m(m+1)(m+2)}{6}$$

$$\sum_{n=1}^{10} \text{trace}(B) = \frac{2 \times 10 \times 11}{2} + \frac{10 \times 11 \times 12}{6} = 330$$

$$\lim_{m \rightarrow \infty} \left(\frac{\sum \text{trace}(B)}{m^3} \right) = \frac{1}{6}$$

5. **Ans (B,D)**

$$D = \begin{vmatrix} a & 1 & 1 \\ 1 & 1 & b \\ c & 1 & 1 \end{vmatrix} \times \begin{vmatrix} 1-b & bc-1 & 1-c \\ 0 & a-c & c-a \\ b-1 & 1-ab & a-1 \end{vmatrix}$$

$$= D_1 D_2$$

D_2 is made by cofactors of D_1 so $D_2 = D_1^2$

$$D = D_1^3$$

$$= \begin{vmatrix} a & 1 & 1 \\ 1 & 1 & b \\ c & 1 & 1 \end{vmatrix}^3 = [(a-c)(1-b)]^3 = 5^3 = 125$$

6. **Ans (A,B,C)**

Here $P^T = -P$

$$Q = (I + P)(I - P)^{-1}$$

$$Q^T = [(I + P)(I - P)^{-1}]^T$$

$$= (I - P^T)^{-1} (I + P^T)$$

$$= (I + P)^{-1} (I - P)$$

$$|Q||Q^T| = 1, |Q| = 1, |2Q| = 2^3|Q| = 8$$

$$|\text{adj}Q| = |Q|^{3-1} = 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

Constant term (absolute term) will be same as $g(0)$

$$\text{since } g(y) = Ay^3 + By^2 + Cy + D$$

$$g(0) = (p+q+r)(p^2+q^2+r^2-pq-qr-rp) > 0$$

8. **Ans (D)**

$$g'(y) = 3Ay^2 + 2By + C$$

$$\Rightarrow C = g'(0)$$

$$g'(0) = C = 3qr - 3p^2$$

9. **Ans (A)**

$$\text{Let } U = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$$\therefore AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \left. \begin{array}{l} a + 2b + 3c = 1 \\ 2b + 3c = 0 \\ 3c = 0 \end{array} \right\}$$

$$\Rightarrow a = 1, b = 0, c = 0$$

$$AU_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \left. \begin{array}{l} d + 2e + 3f = 0 \\ 2e + 3f = 1 \\ 3f = 0 \end{array} \right\}$$

$$\Rightarrow d = -1, e = \frac{1}{2}, f = 0$$

$$AU_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \left. \begin{array}{l} g + 2h + 3i = 0 \\ 2h + 3i = 0 \\ 3i = 1 \end{array} \right\}$$

$$\Rightarrow g = 0, h = -\frac{1}{2}, i = \frac{1}{3}$$

$$\Rightarrow U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

$$|U| = \frac{1}{6}$$

10. Ans (C)

$$\text{Let } U = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$$\therefore AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} a + 2b + 3c = 1 \\ 2b + 3c = 0 \\ 3c = 0 \end{cases}$$

$$\Rightarrow a = 1, b = 0, c = 0$$

$$AU_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} d + 2e + 3f = 0 \\ 2e + 3f = 1 \\ 3f = 0 \end{cases}$$

$$\Rightarrow d = -1, e = \frac{1}{2}, f = 0$$

$$AU_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{cases} g + 2h + 3i = 0 \\ 2h + 3i = 0 \\ 3i = 1 \end{cases}$$

$$\Rightarrow g = 0, h = -\frac{1}{2}, i = \frac{1}{3}$$

$$\Rightarrow U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

$$U^{-1} = \frac{\text{adj}U}{|U|} = \frac{1}{6} \begin{bmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow U^{-1} \text{ is triangular matrix.}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (7.50)

Possible when $P = 8, Q = 7$

and $A + B + C = 7$ ($A, B, C \geq 1$)

number of possible solution is ${}^{7-1}C_{3-1} = 15$

2. Ans (1.20)

Possible when A, B & C all are odd

$$A + B + C = 7$$

$$\text{put } A = 2x - 1, B = 2y - 1 \text{ \& } c = 2z - 1$$

$$\Rightarrow x + y + z = 5 \{x, y, z \geq 1\}$$

$$\Rightarrow {}^{5-1}C_{3-1} = {}^4C_2 = 6 \text{ solutions}$$

3. Ans (56.00)

L_1 divides plane in 2 parts

L_2 in 4 parts

L_3 in 7 parts

L_4 in 11 parts

Let

$$S_n = 2 + 4 + 7 + 11 + \dots + T_n$$

$$S_n = 2 + 4 + 7 + \dots + T_{n-1} + T_n$$

$$0 = [2 + 2 + 3 + 4 + \dots + n \text{ term}] - T_n$$

$$\Rightarrow T_n = 2 + 2 + 3 + 4 + 5 + \dots + n \text{ terms}$$

$$= 1 + \frac{n(n+1)}{2}$$

$$\text{So } T_{10} = 1 + \frac{10 \times 11}{2} = 56$$

4. Ans (5.00)

$$Q_n = \prod_{r=1}^n \frac{1 + \frac{(n+1)(n+2)}{2} - 1}{1 + \frac{n(n+1)}{2} - 1}$$

$$Q_n = \prod_{r=1}^n \left(\frac{n+2}{n} \right)$$

$$Q_n = \frac{3}{1} \times \frac{4}{2} \times \frac{5}{3} \times \frac{6}{4} \times \dots \times \frac{n+1}{n-1} \times \frac{n+2}{n}$$

$$Q_n = \frac{(n+1) \cdot (n+2)}{2}$$

$$\text{Now } \sum_{r=1}^n \frac{1}{Q_r} = 2 \sum_{r=1}^n \frac{1}{(r+1)(r+2)}$$

$$= 2 \sum_{r=1}^n \left(\frac{1}{r+1} - \frac{1}{r+2} \right)$$

5. Ans (0.50)

$$\begin{aligned} \sum_{r=0}^n \frac{r^2}{nCr} &= a_n + \sum_{r=0}^n \frac{(r-1)(r+1)}{nCr} \\ &= a_n + (n+1) \sum_{r=0}^n \frac{r+2-3}{n+1Cr+1} \\ &= a_n + (n+1)(n+2) \sum_{r=0}^n \frac{1}{n+2Cr+2} - 3(n+1) \sum_{r=0}^n \frac{1}{n+1Cr+1} \\ &= a_n + (n+1)(n+2) \left(a_{n+2} - 1 - \frac{1}{n+2} \right) - 3(n+1) \\ &\quad (a_{n+1} - 1) \\ &= a_n + (n+1)(n+2)a_{n+2} - 3(n+1)a_{n+1} - n(n+1) \\ &\Rightarrow P_n = (n+1)(n+2), Q_n = -3(n+1), R_n = -n(n+1) \end{aligned}$$

6. Ans (24.00)

$$\begin{aligned} \sum_{r=0}^n \frac{r^2}{nCr} &= a_n + \sum_{r=0}^n \frac{(r-1)(r+1)}{nCr} \\ &= a_n + (n+1) \sum_{r=0}^n \frac{r+2-3}{n+1Cr+1} \\ &= a_n + (n+1)(n+2) \sum_{r=0}^n \frac{1}{n+2Cr+2} - 3(n+1) \sum_{r=0}^n \frac{1}{n+1Cr+1} \\ &= a_n + (n+1)(n+2) \left(a_{n+2} - 1 - \frac{1}{n+2} \right) - 3(n+1) \\ &\quad (a_{n+1} - 1) \\ &= a_n + (n+1)(n+2)a_{n+2} - 3(n+1)a_{n+1} - n(n+1) \\ &\Rightarrow P_n = (n+1)(n+2), Q_n = -3(n+1), R_n = -n(n+1) \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (9)

$$\begin{aligned} 2A + 3BB^T &= I \quad \dots(1) \\ \text{Now Taking transpose} \\ 2A^T + 3BB^T &= I \quad \dots(2) \\ \text{from (1) \& (2) } A &= A^T \text{ \& } A^T = B^{-1} \text{ (Given)} \\ \Rightarrow A &= B^{-1} \text{ or } A^{-1} = B \quad \dots(3) \\ BA &= BB^{-1} = I \\ \text{Again } A^T &= B^{-1} \\ A^T &= B^{-1} \\ A^TB &= B^{-1}B = I \\ \text{from (3) } B^T &= (A^{-1})^T = (A^T)^{-1} \\ B^T &= (B^{-1})^{-1} = B \\ \text{Now } 2A + 3BB^T &= I \\ 2AB + 3B^3 &= B \Rightarrow 3B^3 = B - 2I \\ \text{so } \alpha &= |A^{-1} - 3B^3 + BA| = |B - (B - 2I) + I| = 27 \\ \beta &= \text{tr}(A^{-1} - 3B^3 + BA) = \text{tr}(B - (B - 2I) - I) = 3 \end{aligned}$$

8. Ans (1)

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} \cdot {}^nC_r \left(\sum_{t=0}^{r-1} {}^rC_t 3^t \right) \\ = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} \cdot {}^nC_r \cdot [4^r - 3^r] \\ = \lim_{n \rightarrow \infty} \frac{5^n - 4^n}{5^n} = 1 \end{aligned}$$

9. Ans (0)

$$(2b - 3c)x + (c - 2a)y + (3a - b)z = 0$$

$$(2c - 3a)x + (a - 2b)y + (3b - c)z = 0$$

$$(2a - 3b)x + (b - 2c)y + (3c - a)z = 0$$

$\Rightarrow$ This can be converted to matrix equation

$$AX = 0$$

$$\text{where } A = \begin{bmatrix} 2b-3c & c-2a & 3a-b \\ 2c-3a & a-2b & 3b-c \\ 2a-3b & b-2c & 3c-a \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2b-3c & c-2a & 3a-b \\ 2c-3a & a-2b & 3b-c \\ 2a-3b & b-2c & 3c-a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{vmatrix} = 0$$

$\therefore$ Skew symmetric matrix of odd order

determinant is zero.

$$\det(A) + \det(2A) + \det(3A) = 0$$